Can use Poisson kernel $P(x,y) = \dfrac{1-|x|^2}{\omega_{n-1}\,|x-y|^n}$

$x \in B$, $y \in \partial B$,

ω_{n-1} = surface area of ∂B

$\Rightarrow$ solution of $(*)$ is

$$f(x) = \int_{\partial B} g(y)\, P(x,y)\, dA_y$$

<u>Maximum principle</u> Let $f : \overline{B} \longrightarrow \mathbb{R}$ be continuous

(weak) s.t. $\Delta f \geq 0$ in B (f is subharmonic)

Then $\sup_B f = \sup_{\partial B} f$

<u>Proof</u> suppose $x_0 \in B$ s.t.

$$f(x_0) > \sup_{\partial B} f$$

Let $g(x) = f(x) + \varepsilon |x - x_0|^2$ for $\varepsilon > 0$

If $\varepsilon > 0$ sufficiently small, then

$$g(x_0) > \sup_{\partial B} g \quad \text{and} \quad \Delta g > 0 \text{ in } B$$

g achieves max at some $x_{max} \in B$

$\Rightarrow$ $\Delta g(x_{max}) \leq 0$ contradicts $\Delta g > 0$ $\square$

E.g. <u>Proposition</u> Suppose that $f = 0$ on ∂B
and $|\Delta f| < \varepsilon$ on B

Then $|f| \leq \frac{\varepsilon}{2n}$ on B

<u>Proof</u> $\Delta f > -\varepsilon$

$$\Rightarrow \Delta\left(f + \frac{\varepsilon}{2n}|x|^2\right) > 0 \qquad \left(\Delta |x|^2 = 2n\right)$$

$f = 0$ on $\partial B \Longrightarrow f + \frac{\varepsilon}{2n}|x|^2 = \frac{\varepsilon}{2n}$ on ∂B

max principle $\Rightarrow$ $\underbrace{f + \frac{\varepsilon}{2n}|x|^2}_{\geq f} \leq \frac{\varepsilon}{2n}$ on B

So $f \leq \frac{\varepsilon}{2n}$ on B.

Same argument applies to $-f$, so $-f \leq \frac{\varepsilon}{2n}$ on B

i.e. $|f| \leq \frac{\varepsilon}{2n}$ on B $\square$

<u>Proposition</u> (Interior estimates) Let $f : B \to \mathbb{R}$ be harmonic ($\Delta f = 0$). There are constants $C_k > 0$ s.t.

$$\sup_{B(0,1/2)} |\nabla^k f| \leq C_k \sup_{B(0,1)} |f|$$

<u>Proof</u> e.g. use maximum principle
or differentiate Poisson formula $\square$

Schauder estimates

Theorem Let $\Delta f = g$ in B_1 $\left(\begin{array}{l}\text{use notation}\\ B_r = B(0,r)\end{array}\right)$

Then $\|f\|_{C^{2,\alpha}(B_{1/2})} \leq C\left(\|g\|_{C^\alpha(B_1)} + \|f\|_{C^0(B_1)}\right)$

for a constant C (depending on n, α)

Proof Enough to construct a good quadratic approx. to f at the origin. I.e. want quadratic P s.t. $\sup_{B_r} |f - P| \leq C r^{2+\alpha}$ for $r < \frac{1}{2}$

Replace f by $f - \frac{1}{2n} g(0)\,|x|^2$

$\Rightarrow$ can assume $g(0) = 0$

Divide both sides by $\sup_{B_1} |f| + \varepsilon^{-1}\|g\|_{C^\alpha(B_1)}$

with ε to be chosen

$\Rightarrow$ can assume $|f| \leq 1$ on B_1, $\|g\|_{C^\alpha} \leq \varepsilon$ on B_1

Let h solve $\begin{array}{l}\Delta h = 0 \text{ in } B_1\\ h = f \text{ on } \partial B_1\end{array}$ $(\Delta f = g)$

$\begin{array}{l}\Delta(f-h) = g \text{ in } B_1\\ f - h = 0 \text{ on } \partial B_1\end{array}$, $|g| \leq \varepsilon$ on B_1

max principle $\Rightarrow$ $|f - h| \leq \frac{\varepsilon}{2n}$ on B

Let $P_1 = 2^{nd}$ order Taylor expansion of h at 0

interior estimates $\Rightarrow$ $|h - P_1| \le C r^3$ on B_r for $r < \frac{1}{2}$

$\Rightarrow$ $|f - P_1| \le C(r^3 + \varepsilon)$ on B_r for $r < \frac{1}{2}$

choose r_0 small s.t. $C r_0^3 < \frac{1}{2} r_0^{2+\alpha}$

then choose ε s.t. $C\varepsilon < \frac{1}{2} r_0^{2+\alpha}$

$\Rightarrow$ $|f - P_1| \le r_0^{2+\alpha}$ on B_{r_0}

<u>Rescale</u>: Let $f_1(x) = \frac{1}{r_0^{2+\alpha}} (f - P_1)(r_0 x)$

then $|f_1| \le 1$ on B_1

$$\Delta f_1(x) = \frac{1}{r_0^{2+\alpha}} r_0^2 \Delta f(r_0 x) = \frac{1}{r_0^{\alpha}} g(r_0 x) = g_1(x)$$

$$\|g_1\|_{C^\alpha} \le \varepsilon$$

Can repeat argument, get P_2 s.t.

$$|f_1 - P_2| \le r_0^{2+\alpha} \text{ on } B_{r_0}$$

Scale back:

$$|f(x) - P_1(x) - r_0^{2+\alpha} P_2(x/r_0)| \le (r_0^2)^{2+\alpha} \text{ on } B_{r_0^2}$$

Iterate this procedure $\rightsquigarrow$ sequence P_k

can define $P(x) = P_1(x) + r_0^{2+\alpha} P_2(x/r_0)$

$$+ (r_0^2)^{2+\alpha} P_3(x/r_0^2) + \dots$$

(converges since coeffs. are geometric series)

P has bounded coefficients and

$$|f - P| \le C(r_0^k)^{2+\alpha} \quad \text{on } B_{r_0^k}$$

$k = 1, 2, 3, \dots$

If $r_0^{k+1} \le r \le r_0^k$ then $|f - P| \le C(r_0^k)^{2+\alpha}$ on B_r

$r_0^k \le r/r_0$

$$\le C r_0^{-(2+\alpha)} r^{2+\alpha} \quad \square$$